A Vision of System-level Adaptation in Digital Twin Networks

Prasad Talasila*, Riccardo Sieve‡, Roberto Casadei†, Samuele Burattini†, Lukas Esterle*
* Department of ECE, Aarhus University, Aarhus, Denmark
† Department of Computer Science and Engineering, University of Bologna, Cesena, Italy
‡ Department of Informatics, University of Oslo, Oslo, Norway

***Abstract*—Digital Twins are increasingly deployed as interconnected systems rather than isolated entities, forming networks that collectively support the engineering of complex cyber-physical applications. While adaptation has been studied for individual Digital Twins, enabling coordinated adaptation across Digital Twin networks remains an open challenge. This paper presents a vision for adaptation in networked Digital Twin systems. We propose a vision, a reference architecture, and identify the key challenges of self-adaptive Digital Twin networks to support adaptive coordination. Finally, we discuss open research directions and outline some challenges to be tackled to make this vision a reality.**

***Index Terms*—Digital Twin Networks, Self-adaptation, Orchestration, Cyber-physical ecosystems**

I. Introduction

Digital Twins (DTs) [1] are virtual (synchronised) representations of physical systems, the *Physical Twins (PT)*, that can be used to monitor, analyse, and optimise the performance of the PTs they represent. DTs are increasingly used in a wide range of *collective cyber-physical system (CPS)* applications [2] spanning domains such as manufacturing, healthcare, transportation, and smart cities. However, as the PTs that DTs represent are often complex and dynamic, DTs must adapt to changing conditions to remain accurate and useful over time, even amid the uncertainty and dynamics of the real world. This can be achieved through the use of machine learning (ML) algorithms [3] and artificial intelligence [4] that can learn from data and adapt to changing conditions. Additionally, DTs can be designed to be modular and flexible [**?**], allowing for easy updates [5] and modifications [6] as needed, as well as to evolve with changing conditions [7], [8], or with multi-dimensional models [9]. We can distinguish adaptation into structural [10] and behavioural [11] adaptation, where the former focuses on the physical structure of the PT, thereby requiring an adaptation of the DT as well. The latter focuses on the behaviour and functionality of the DT, for example when it needs to change its strategy and policies based on changing environmental conditions.

DT adaptation can happen at two levels. The first level is adaptation of a single DT, which is more appropriate for a stand-alone DT paired with a PT. DTs can further adapt their behaviour by identifying the different states of the system they represent [12], and adjusting their behaviour to apply appropriate strategies [13]. However, DT-based systems usually do not operate in isolation in the real world; they interact with other systems throughout their lifetime. This requires DTs to self-integrate and self-optimise in such situations [14]. It also creates a network of DTs that are dynamically integrated and removed at runtime. Thus, the second-level is a coordinated adaptation among a network of DTs. The DTs of Internet routers, Radio Access Networks (RANs) of 6G cellular networks, and factory production systems are prime examples of networks of DTs. Adaptation in such systems requires coordination among the DTs within the network. This coordination can either be distributed among the DTs themselves, i.e. decentralised adaptation, or coordinated by a single Manager, i.e. centralised adaptation. The degree of autonomy and capabilities of DTs determines the suitability of decentralised and centralised adaptation. When dealing with managed adaptation, the interactions can be aided by an orchestrator [15], which can facilitate coordination and collaboration for real-time optimisation, potentially through the use of ML or multi-agent systems [16]–[18].

In this paper, we propose a vision for ***self-adaptive DT networks*** (SA-DTN), where the DTs can adapt themselves and their interactions with other DTs in the network to achieve a desired system state. We discuss the challenges and opportunities of managed and self-adaptation of networked DTs, and propose a reference architecture for enabling SA-DTN. We also discuss the implications of self-organisation for the design and implementation of DTs, and identify future research directions for this emerging area.

II. System Model and Problem Statement

We consider a networked DT system consisting of a set of PTs, each represented by a corresponding DT and deployed on a shared computing infrastructure. Formally, we model the system as

$$\mathcal{N} = (P, D, R, \mathbb{I}, \mathbb{C}),$$

where:

- $P = \{p_1, \ldots, p_n\}$ is the set of PTs;
- $D = \{d_1, \ldots, d_n\}$ is the corresponding set of DTs;
- $R \subseteq D \times D$ is the set of relationships between DTs;
- $\mathbb{I}$ denotes the communication infrastructure through which DTs exchange information;
- $\mathbb{C}$ represents the shared computing infrastructure that determines the resources available to the DT network (DTN).

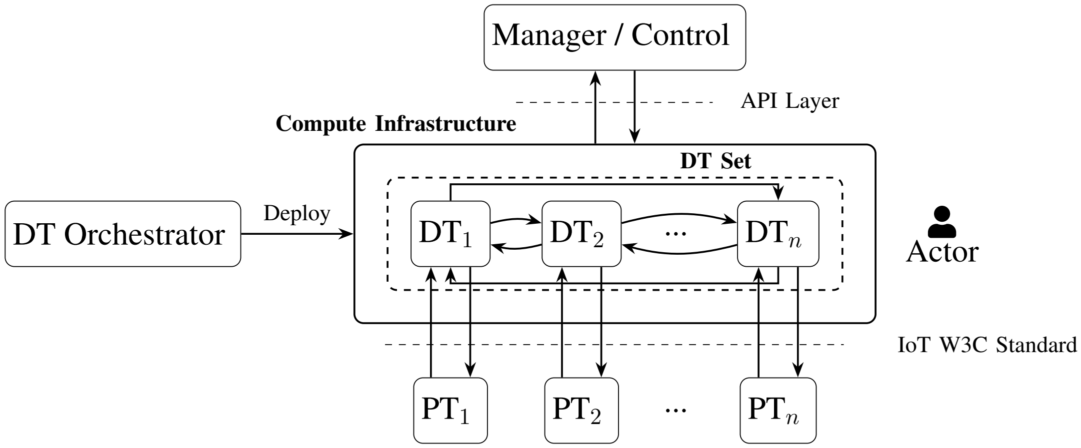

Fig. 1. A high-level reference architecture of a Digital Twin network ($\mathcal{N}$). The figure illustrates the relationships (R) between the Orchestrator, the DTs (D), the PTs (P), and the Manager/Control. The DTs are placed in compute ($\mathbb{C}$) and communication ($\mathbb{I}$) infrastructure.

The model is depicted in Figure 1, showing multiple DTs, each independently mirroring an individual PT, deployed on the same computing infrastructure. The figure includes a DT orchestrator and a Manager component, whose roles are discussed in Section III and are external to $\mathcal{N}$.

For simplicity, although there might be systems in which the same asset is mirrored by multiple DTs, we assume in this system model that each DT is associated with exactly one PT, and vice versa, with no loss of generality. Relationships in R capture dependencies and interactions among DTs that mirror those occurring in the physical world. We assume all DTs expose a common communication interface, allowing DTs to dynamically join and leave the network while remaining interoperable with existing DTs. Furthermore, DTs compete for the computational resources provided by the shared infrastructure C. This requirement motivates the need for adaptation at the network level, as the adaptation of an individual DT may indirectly affect the behaviour of other DTs, e.g. by requiring additional resources and thus degrading the overall Quality of Service (QoS) of the DTN and the timeliness of DT-PT interactions, ultimately affecting the controllability of the system of PTs.

We further characterise each DT as:

$$d_i = (M_i, \mathbf{s}_i, K_i, \alpha_i),$$

where M_i is the set of digital models the DT uses to represent the PT, $\mathbf{s}_i$ is the sequence of states $(s_j, ..., s_n)$ computed by the DT over time, K_i is the set of controls that the DT can send to the PT to influence its behaviour towards achieving a desired state, and α_i represents its adaptation policy that controls how the DT can either change its current model, or control the PT to reach a desired state. A model $m \in M_i$ is the component of the DT responsible for interpreting the inputs from the PT and producing a new element s_x in the DT state sequence $\mathbf{s}_i$ as the output. The state represents the properties and conditions of the PT in soft real-time, and can be used by applications interested in monitoring the PT. Reacting to state changes may trigger the adaptation policy α_i to either change the model with a different $m' \in M_i$ or send a control $c \in K_i$ directed at the PT to achieve a desired state.

This simple model captures the essence of a DT, as a software component that models a PT, memorises and monitors its state over time, and influences its behaviour through a bi-directional data flow.

A. Problem Statement

In this paper, given a networked DT system ($\mathcal{N}$), we are interested in exploring the challenges of enabling the DTs to operate in a network to adapt to dynamic environment affecting the PTs (P), the set of DTs (D), relationships between DTs (R) or the available computational resources ($\mathbb{C}$). Examples of this could be changing traffic conditions in DT-enabled connected autonomous vehicles, failing or changing conditions in the PT-DT communication, or updated goals in distributed search-and-rescue operations. For the scope of this paper, we thus assume that the communication infrastructure between DTs of the network $\mathbb{I}$ is not considered a source of potential adaptation and that DTs are always able to exchange information among themselves.

Such adaptation should allow the network to achieve desired system-wide objectives while preserving the autonomy of individual DTs. We are interested in supporting individual DT-level adaptation that reacts to local conditions and requirements of the DT-PT interaction, thereby enabling *passive* adaptation of the network to changing conditions. We are also interested in *managed* adaptation of the whole network to accomplish system-level goals, such as supporting coordination amongst DTs to meet a desired level of service.

As we elaborate in the next section, the key challenge is that DT adaptations are inherently local—they are initiated or executed by individual DTs—yet their effects may propagate through the network via inter-DT dependencies. Consequently, local adaptation decisions may interfere with one another, compromise global objectives, or create emergent behaviours that cannot be understood by considering DTs in isolation.

This paper, therefore, investigates the adaptation capabilities required by individual DTs, distinguishes between centralised and decentralised adaptation, and identifies the mechanisms needed to coordinate these adaptations such that the DTN collectively converges towards the desired system state.

III. Solutions for Adaptive Self-Organisation

In this section, we describe the solutions for adaptation, both for a single DT node, and for a DTN.

A. Adaptation for Single Digital Twin

The adaptation process in a networked DT system ($\mathcal{N}$) can be defined as a feedback loop, where each DT (d_i) continuously updates its model and behaviour based on the updates it receives from the corresponding PTs (p_i) and the other DTs ($d_j \in D \setminus \{d_i\}$) in the network. This feedback loop allows the DTs to learn and improve over time, making them more effective at representing the PTs and providing useful insights. We consider the adaptation process to follow the MAPE-K loop [19], where a DT goes through four main steps: Monitor, Analyse, Plan, and Execute, with a shared Knowledge base that informs the adaptation process.

Within the MAPE-K loop, the different steps can reflect different aspects of the adaptation process. The Monitor step involves a DT (d_i) collecting data from its linked PT (p_i) and other DTs ($d_j \in D \setminus \{d_i\}$) in the networked DT system ($\mathcal{N}$). This data concerns the state of the system and the model. The Analyse step involves analysing the data collected in the Monitor step to identify patterns, trends, and anomalies that can inform the adaptation of the model and the state. The Plan step involves developing a plan to adapt the state (s_i) of the DT based on the insights gained from the Analyse step. Finally, the Execute step involves implementing the plan developed in the Plan step, allowing the DT to adapt its behaviour. This is not the control of the PT (i.e., c_i) but a dedicated control step for the adaptation of the DT. The MAPE-K loop can also be used as an adaptation policy (α_i), where the different steps of the loop are used to inform the adaptation decisions of the DTs. The Knowledge base can therefore allow the DTs to learn and improve over time.

The models (M_i) of the DT can be fine-grained or coarse-grained. The choice of model granularity depends on the specific requirements of the application and the characteristics of the PT being modelled. Fine-grained models can provide more detailed insights and allow for more precise adaptation, but they may also require more computational resources and may be more complex to manage. Coarse-grained models, on the other hand, may be less detailed but can be more efficient and easier to manage. The adaptation process can be applied to both types of models, allowing for flexibility in how the DTs are designed and implemented. The adaptation process can also be designed to leverage machine learning algorithms, rule-based systems, or a combination of both. A multi-fidelity model has been proposed [20], allowing a DT to switch between different fidelities of the same model. This allows the DT to select the appropriate model given the required granularity and the available computational resources. Historically, DTs have been specified and deployed on cloud compute environments using standards such as the *DT Definition Language* [21]. Such standards can form a basis for a standardised approach to adaptation that can be applied across different DT implementations. This can help to ensure that the adaptation process is consistent and effective, regardless of the specific implementation of the DTs.

B. Adaptation for the Digital Twin Network

Within a DTN, like the one shown in Fig. 1, the adaptation process can be divided into two main (possibly concurrent) components: centralised and decentralised. The *centralised* approach involves a central manager that coordinates the adaptation of the DTs, while the *decentralised* approach involves the DTs coordinating their adaptation through peer-to-peer communication. Specifically, in the centralised approach, the central manager collects data from the DTs, analyses it, and develops a plan for adapting the behaviour of the DTs. In the decentralised approach, the DTs communicate with each other through communication infrastructure ($\mathbb{I}$) to share data and coordinate their adaptation, allowing them to adapt their behaviour based on the collective knowledge of the network.

The "triggers" for adaptation can be *endogenous*, originating from within the DT itself, or *exogenous*, where the manager takes responsibility for adaptation if the DT does not have this capability. The manager needs to make decisions on which DTs to adapt, and how to adapt them, based on the data it receives from DTs and PTs. The process can be guided by a shared policy or database for coordination of adaptation, where the manager and the DTs can access and update this shared policy or database to inform their adaptation decisions. The choice of manager-centric or DT-centric adaptation depends on the specific characteristics of the DT network, the underlying infrastructure, and the PTs being modelled.

Furthermore, if a single DT is incapable of self-adaptation, the manager can take over the responsibility for adaptation, ensuring that the overall system continues to function effectively. This allows for a better separation of adaptation responsibilities. Moreover, we can also consider a *hybrid* approach, where the manager and the DTs share the responsibility for adaptation of the entire system, whether ensuring correctness of operational DTs or adding / removing them to represent the state of the PT. In this case, the manager can provide high-level guidance and coordination, while the DTs can take care of the low-level adaptation based on their local knowledge and data. This approach allows for a more flexible and scalable adaptation process, as it can leverage the strengths of both centralised and decentralised approaches.

The DTs can communicate with each other or be connected through a central manager over standard data and protocol interfaces (W3C standards / RFCs). The adaptation process can be designed to leverage this communication and coordination, allowing the DTs to share data and insights, and to adapt their behaviour based on the collective knowledge of the network. This can help to ensure that the overall system continues to function effectively, even as the individual DTs adapt and evolve over time. The network of DTs with standard interfaces can also be considered as an open system where different DTs can join and leave the network, and the adaptation process can be designed to handle these changes. This allows for a more dynamic and flexible system that can adapt to changes in the network and the PTs being modelled.

Finally, the manager needs to accommodate diverging requirements of participating DTs in the adaptation process. For example, some DT applications may require real-time adaptation, while others may allow for more gradual adaptation over time. The adaptation process can be designed to be flexible, allowing it to accommodate a wide range of application requirements and constraints, ensuring that the overall system continues to function effectively while meeting the specific needs of the applications being modelled.

C. DT Orchestrator

The DT orchestrator shown in Fig. 1 is responsible for the deployment of DTs at runtime and keeping them up-to-date when PTs are either added or removed (i.e., adding or removing the corresponding DTs from the DTN). The orchestrator plays a key role in situations where negotiation for, and orderly allocation of limited resources among DTs is necessary. The DT continuum [22] has proven to be a suitable deployment technique for DTs, and such distributed deployment requires an active DT orchestrator. Another situation demanding the services of the DT orchestrator is the handling of multi-fidelity models in DTs. Because running DTs at maximum fidelity at all times may require substantial resources, the adaptation process can be designed to adjust the fidelity of DTs based on current needs and constraints of a DTN. This allows for a more efficient use of resources, while still providing sufficient insights and information about the PTs being modelled. The adaptation process can be designed to dynamically adjust the fidelity of the DTs based on factors such as the current workload, the importance of the insights being generated, and the available resources, ensuring that the overall system continues to function effectively while managing resource constraints.

IV. Related Work

A. Digital Twin Network

The term DTN refers to a virtual, synchronised representation of a network of PTs. However, multiple interpretations exist. Some authors [23] and recommendations like *ITU-T Y.3090* consider a DTN as a virtual representation of a telecommunication network. This paper adopts the existing interpretation of DTN as an ensemble of interconnected DTs that collaborate and share information to address complex tasks [24]. Concrete DTN instantiations may exist in different domains, including smart manufacturing, vehicular systems, and smart cities.

Typically, DTNs are envisioned as systems deployed by a single organisational entity with the purpose of supporting the collaboration between DTs in the system [24]. Accordingly, in our system model (Section II), we include the computing infrastructure as part of the network, as it is expected that the stakeholders managing the network can also manage how DTs are deployed.

Among the main functionalities of DTNs, a prior survey [25] highlights anomaly monitoring, system state estimation, resource allocation, task offloading, model optimisation, and security, focusing on ML-driven approaches. Resource allocation and task offloading are crucial for DTNs, which are used to monitor and simulate the global state of computing and communication resources in real time. The survey describes deep-learning techniques for deriving optimal offloading and allocation strategies that jointly minimise latency, energy consumption, and QoS degradation. These techniques can be embedded in the manager component proposed in this work.

A related, emerging concept in the DT literature is that of DT *ecosystems* [26], which focus on the relationships among several independently deployed DTs at a semantic level. These ecosystems are further characterised by an interface that allows external consumers to seamlessly interact with DTs and dynamically discover their relationships. Our proposal for networked DT systems envisions a similar interface directed towards a manager component that can coordinate adaptation of the network by being aware of the state and relationships of DTs in the system. In contrast, however, DT ecosystems typically consider mash-ups of independently managed DTs with no common orchestration.

B. Management, Adaptation, and Self-Adaptation in Digital Twin Networks

Nguyen et al. [15] address the topic of *orchestration* of multiple DTs through a framework based on four key functions: federation, translation, brokering, and synchronisation. These functions respectively cover aggregation, interoperability, communication, and digital-physical alignment for multiple DTs. Regarding adaptation and DTNs, existing work mostly focuses on adaptation in communication network DTs (cf. [27]), which is unrelated to the focus of this paper.

Previous works on self-integration and self-organisation in DTs focus instead on leveraging DTs as abstractions to coordinate the PTs. The proposal of *collective* DT [28] applies the DT concept to provide a synchronised virtual representation of an entire group of PTs (e.g., a human crowd [29]), possibly but not necessarily using an ensemble of individual DTs. A different perspective is adopted in [30], which discusses the role of DTs in supporting self-integration by enabling the exchange of models for PTs or allowing DTs to request services from other DTs at runtime.

To the best of our knowledge, no contributions address self-adaptation at the DT network level. Such self-adaptation would support the dynamic reconfiguration of DTs on shared infrastructure in response to external conditions. Our work aims to provide preliminary ideas to fill this gap. Unlike previous work, the focus is not on leveraging DTs to improve the self-* properties of the target system of PTs, but rather on having DTs themselves adapt in order to maintain a desired QoS for the entire networked system.

V. Future Challenges

Complex CPS, connected into large systems (cf. [29]), face many challenges. Incorporating the idea of networked DTs would support the deployment, operation, and continuous improvement of such systems. Specifically, monitoring and (potentially autonomous) control leveraging the presented ideas of networked DTs can be realised. However, dedicated challenges arise that need to be tackled in order to make this idea a reality.

- **Security:** Networked DTs, enabled to directly interact and interoperate, require a heightened level of security. This includes security against both external malevolent parties and the introduction of malevolent DTs into the network. Dedicated techniques that also consider the *collective* nature of DTNs [31] are required that can ensure such security even among autonomously operating DTs.
- **Trustworthiness and correctness:** Enabling systems to rely safely on another system's output within the network requires techniques like runtime verification and self-organising trust [32]. While the *Manager* component can support such activities, enabling DTs to distribute the issue across the network would remove a single point of computation and potential failure.
- **Robustness and Synchronisation in the DTN:** Changes introduced into DTs by linked PTs must be propagated to the DTN, meaning that communication reliability plays a crucial role. The DTs belonging to a DT set are connected over a potentially volatile network, which introduces the problems related to intermittency, variable latency and QoS. "In this context, centralised adaptation faces more challenges than decentralised adaptation, because the manager in a centralised approach becomes a single point of failure for the system. The possible large-scale nature of DTNs can exacerbate the "synchronisation" problem, namely the propagation of changes between individual PTs and DTs [33], due to increased load on the network.
- **Migrating from IoT systems to DTNs:** As IoT systems become increasingly prevalent, establishing and deploying the corresponding DT networks for legacy systems becomes increasingly challenging. Dedicated tools for mass-deployment in large-scale, physically distributed, and heterogeneous networks are required.
- **Semantic interoperability:** Establishing a DTN requires participating DTs to be compatible with each other. This encompasses both sharing information and interacting and co-simulating together. This can both be accomplished through mutual and clearly defined interfaces. In addition, DTs require approaches to exchange their underlying semantics for a meaningful exchange, specifically when interacting with each other. Approaches for semantic understanding in DTs are being investigated [34]. However, general solutions for DTNs are not yet available.
- **Self-improving integration within DTNs**: Over time, DTNs must not only interact with each other but also self-improve at the network level. This can be addressed by supporting management techniques for *open* ensembles of DTs. For such self-improvement, both local and global constraints and goals should be addressed continuously as they vary according to owner and user needs [14].
- **Runtime Changes in Adaptation Policies**: Defining the strategies that the DTN can use to switch between managed and decentralised adaptation is a challenge for DTNs in which heterogeneous DTs may exhibit different adaptation levels and needs.

VI. Conclusion

In this paper, we present a vision towards self-adaptive Digital Twin Networks (DTNs) to capture networks of Digital Twin (DT)-enabled Cyber-Physical Systems (CPS). This vision highlights the DT characteristics that make adaptation a collective, not individual, problem. We discuss the role of individual and global goals in DTN adaptation and present an abstract framework to integrate them both, envisioning different styles of adaptation by means of centralised coordination via manager and orchestrator components, or with fully decentralised approaches. With the rise of interacting and collaborating CPS that support each other to achieve common goals, this initial vision can support future development of DTNs to further enable self-improvement and self-integration. Specifically for networked DT systems, we outline challenges that require future work to deliver the vision of self-adaptive DTNs. We also highlight the need to address security, semantic interoperability, autonomous trustworthiness, change propagation, migration of existing systems, and self-improving integration [14] as relevant future research directions for DTNs.

Acknowledgements

This publication is related to activities carried out within the project FoMaSE (Foundations for Macro-programming-based Software Engineering, Grant No. FIS-2024-00174, CUP J53C25002170001), funded by the Italian Ministry of University and Research (MUR) under the Italian Science Fund (FIS3) Starting Grant. The work was further supported by the DT-CORE project funded by the Poul Due Jensen Foundation.

References

[1] S. Mihai, M. Yaqoob, D. V. Hung, W. Davis, P. Towakel, and M. Raza, "Digital Twins: A Survey on Enabling Technologies, Challenges, Trends and Future Prospects," *IEEE Commun. Surv. Tutorials*, vol. 24, no. 4, pp. 2255–2291, Sep. 2022.

[2] R. Casadei, L. Esterle, R. F. Gamble, P. Harvey, and E. F. Wanner, "Editorial: Understanding and engineering cyber-physical collectives," *Frontiers Robotics AI*, vol. 11, 2024. [Online]. Available: https://doi.org/10.3389/frobt.2024.1407421

[3] T. Wu, F. Yang, U. Farooq, X. Li, and J. Jiang, "An online learning method for constructing self-update digital twin model of power transformer temperature prediction," *Applied Thermal Engineering*, vol. 237, p. 121728, Jan. 2024.

[4] E. Bellini, F. Bagnoli, M. Caporuscio, E. Damiani, F. Flammini, I. Linkov, P. Liò, and S. Marrone, "Resilience learning through self adaptation in digital twins of human-cyber-physical systems," in *IEEE International Conference on Cyber Security and Resilience, CSR 2021, Rhodes, Greece, July 26-28, 2021*. IEEE, 2021, pp. 168–173.

[5] H. Ejersbo, K. Lausdahl, M. Frasheri, and L. Esterle, "Dynamic Runtime Integration of New Models in Digital Twins," in *2023 IEEE/ACM 18th Symposium on Software Engineering for Adaptive and Self-Managing Systems (SEAMS)*, 2023, pp. 44–55.

[6] D. D. Eneyew, M. A. Capretz, and G. T. Bitsuamlak, "Continuous model calibration framework for smart-building digital twin: A generative model-based approach," *Applied Energy*, vol. 375, p. 124080, Dec. 2024.

[7] Y. Tang, P. Sajadi, M. R. Dehaghani, and G. G. Wang, "A systematic online update method for reduced-order-model-based digital twin," *J. Intell. Manuf.*, vol. 36, no. 8, pp. 5495–5523, 2025.

[8] R.-P. Nikula, M. Paavola, M. Ruusunen, and J. Keski-Rahkonen, "Towards online adaptation of digital twins," *Open Engineering*, vol. 10, no. 1, p. 776–783, Jan. 2020.

[9] H. Zhang, Q. Qi, W. Ji, and F. Tao, "An update method for digital twin multi-dimension models," *Robotics Comput. Integr. Manuf.*, vol. 80, p. 102481, 2023.

[10] E. Kamburjan, R. Sieve, C. P. Baramashetru, M. Amato, G. Barmina, E. Occhipinti, and E. B. Johnsen, "GreenhouseDT: An Exemplar for Digital Twins," in *SEAMS*. ACM, 2024, pp. 175–181.

[11] R. Sieve, E. Kamburjan, F. Damiani, and E. B. Johnsen, "Declarative dynamic object reclassification," in *ECOOP*, ser. LIPIcs. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2025, pp. 29:1–29:31.

[12] R. Sieve, P. Kobialka, A. Pferscher, N. Bencomo, L. S. T. Tarifa, B. S. Rasmussen, and E. B. Johnsen, "A Self-Adaptive Digital Twin Architecture for Dynamic Resource Management," in *SEAMS*. ACM, 2026, pp. –, to appear.

[13] R. Sieve, P. Talasila, P. G. Larsen, and E. B. Johnsen, "A Self-Adaptive Digital Twin Architecture to Automate Greenhouse Management," in *9th IEEE International Conference on Industrial Cyber-Physical Systems, ICPS 2026, Perth, Australia, May 11-14, 2026*. IEEE, 2026, pp. 1–6.

[14] K. Bellman, J. Botev, A. Diaconescu, L. Esterle, C. Gruhl, C. Landauer, P. R. Lewis, P. R. Nelson, E. Pournaras, A. Stein, and S. Tomforde, "Self-improving system integration: Mastering continuous change," *Future Generation Computer Systems*, vol. 117, pp. 29–46, 2021.

[15] D. Nguyen, M. Dao, and K. Zettsu, "Digital Twin Orchestration: Framework and Smart City Applications," in *The Second Workshop on AI for Digital Twins and Cyber-Physical Applications in conjunction with 33rd International Joint Conference on Artificial Intelligence (IJCAI 2024), Jeju - South Korea, August 3rd 2024*, ser. CEUR Workshop Proceedings, vol. 3807. CEUR-WS.org, 2024, pp. 21–40.

[16] D. Li and A. Li, "Cloud-Native Digital Twin Orchestration for Real-Time Decision Optimization Using Fuzzy Constraints and Reinforcement Learning," in *Proceedings of the 2025 ACM Symposium on Cloud Computing, SoCC 2025, Online, USA, November 19-21, 2025*. ACM, 2025, pp. 163–169.

[17] Q. Nie, D. Tang, C. Liu, L. Wang, and J. Song, "A multi-agent and cloud-edge orchestration framework of digital twin for distributed production control," *Robotics Comput. Integr. Manuf.*, vol. 82, p. 102543, 2023.

[18] P. Nguyen, A. Rauniyar, J. Bartel, J. Laufer, C. Dalamagkas, and K. Pohl, "AI-driven digital twin-based security orchestration, automation and response for critical infrastructures," *Autom. Softw. Eng.*, vol. 33, no. 2, p. 61, 2026.

[19] H. Feng, C. Gomes, S. Gil, P. H. Mikkelsen, D. Tola, P. G. Larsen, M. Sandberg, and R. Rezaee, "Integration Of The Mape-K Loop In Digital Twins," in *Annual Modeling and Simulation Conference, ANNSIM 2022, San Diego, CA, USA, July 18-20, 2022*. IEEE, 2022, pp. 102–113.

[20] M. Frasheri, P. Talasila, P. G. Larsen, and L. Esterle, "Towards multi-fidelity digital twins for dynamic adaption and efficient distributed simulation," in *2025 IEEE International Conference on Pervasive Computing and Communications Workshops and other Affiliated Events (PerCom Workshops)*, 2025, pp. 128–133.

[21] C. Schmidt, F. Volz, L. Stojanovic, and G. Sutschet, "Increasing Interoperability between Digital Twin Standards and Specifications: Transformation of DTDL to AAS," *Sensors*, vol. 23, no. 18, p. 7742, 2023.

[22] A. Barbone, S. Burattini, M. Martinelli, M. Picone, A. Ricci, and A. Virdis, "Digital Twin Continuum: a Key Enabler for Pervasive Cyber-Physical Environments," in *2024 33rd International Conference on Computer Communications and Networks (ICCCN)*. IEEE, pp. 29–31.

[23] L. Hui, M. Wang, L. Zhang, L. Lu, and Y. Cui, "Digital Twin for Networking: A Data-Driven Performance Modeling Perspective," *IEEE Network*, vol. 37, no. 3, p. 202–209, May 2023. [Online]. Available: http://dx.doi.org/10.1109/MNET.119.2200080

[24] Y. Wu, K. Zhang, and Y. Zhang, "Digital Twin Networks: A Survey," *IEEE IoT J.*, vol. 8, no. 18, pp. 13 789–13 804, May 2021.

[25] B. Qin, H. Pan, Y. Dai, X. Si, X. Huang, and C. Yuen, "Machine and Deep Learning for Digital Twin Networks: A Survey," *IEEE IoT J.*, vol. 11, no. 21, pp. 34 694–34 716, Jun. 2024.

[26] A. Giulianelli, S. Burattini, A. Ciortea, and A. Ricci, "Hypermedia-based digital twin ecosystems: A proposal for digital twin interoperability," *Future Gener. Comput. Syst.*, vol. 177, p. 108219, Apr. 2026.

[27] Y. Ma, F. Song, G. Pau, I. You, and H. Zhang, "Adaptive Service Provisioning for Dynamic Resource Allocation in Network Digital Twin," *IEEE Network*, vol. 38, no. 1, p. 61–68, Jan. 2024. [Online]. Available: http://dx.doi.org/10.1109/MNET.2023.3337245

[28] R. Casadei, D. Pianini, M. Viroli, and D. Weyns, "Digital Twins, Virtual Devices, and Augmentations for Self-Organising Cyber-Physical Collectives," *Appl. Sci.*, vol. 12, no. 1, p. 349, Dec. 2021.

[29] R. Casadei, G. Delnevo, R. Girau, and S. Mirri, "Crowd digital twins: Motivation, architecture, and roadmap," *Future Gener. Comput. Syst.*, vol. 182, p. 108475, 2026. [Online]. Available: https://doi.org/10.1016/j.future.2026.108475

[30] L. Esterle, C. Gomes, M. Frasheri, H. Ejersbo, S. Tomforde, and P. G. Larsen, "Digital twins for collaboration and self-integration," in *2021 IEEE International Conference on Autonomic Computing and Self-Organizing Systems Companion (ACSOS-C)*. IEEE, pp. 2021–01.

[31] D. Pianini, R. Casadei, and M. Viroli, "Security in Collective Adaptive Systems: A Roadmap," in *IEEE 4th International Workshops on Foundations and Applications of Self* Systems, FAS*W@SASO/ICCAC 2019, Umea, Sweden, June 16-20, 2019*. IEEE, 2019, pp. 86–91. [Online]. Available: https://doi.org/10.1109/FAS-W.2019.00034

[32] R. Casadei, A. Aldini, and M. Viroli, "Towards attack-resistant Aggregate Computing using trust mechanisms," *Sci. Comput. Program.*, vol. 167, pp. 114–137, 2018. [Online]. Available: https://doi.org/10.1016/j.scico.2018.07.006

[33] W. Liu, Y. Fu, Y. Guo, F. Lee Wang, W. Sun, and Y. Zhang, "Two-Timescale Synchronization and Migration for Digital Twin Networks: A Multi-Agent Deep Reinforcement Learning Approach," *IEEE Transactions on Wireless Communications*, vol. 23, no. 11, p. 17294–17309, Nov. 2024.

[34] Y. Huang, S. Dhouib, L. P. Medinacelli, and J. Malenfant, "Semantic Interoperability of Digital Twins: Ontology-based Capability Checking in AAS Modeling Framework," in *2023 IEEE 6th International Conference on Industrial Cyber-Physical Systems (ICPS)*. IEEE, May 2023, p. 1–8. [Online]. Available: http://dx.doi.org/10.1109/ICPS58381.2023.10128003